



PROG7311 Part 2

Activity: Formative 2 Part 2

Course: Programming 3A PROG7311 2026 FT BCAD0701 EMGPMD Term1 GR01

Name: Marco Marais

Criteria	Does not meet the required standard	Meets the required standard	Partially exceeds the required standard	Greatly exceeds the required standard	Criterion Score
1. Database Architecture (EF Core) Schema, Keys, Connections.	2 points Database crashes or is missing. Hardcoded connection strings.	5 points - EF Core configured - Tables exist but lack relationships. 	2 points Schema is correct (Client, Contract, Request). One-to-Many relationships work.	10 points - Perfectly normalised schema. - Fluent API or Attributes used for constraints. - Connection string in appsettings.json.	5 / 10
Criterion Feedback No migration script provided					

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2. Design Pattern Implementation (From Part 1) Translating UML to Code.	4 points - No patterns implemented. - Controllers call DbContext directly - Logic is tightly coupled.	5 points Very basic implementation of design patterns used	14 points Moderate implementation of design pattern	20 points Complete implementation of design patterns used. 	20 / 20
3. Workflow & Validation Logic Business Rules (Status checks).	4 points No validation. "Expired" contracts allow requests.	7 points - Basic validation (Required fields). - Workflow logic has bugs. 	11 points Correctly prevents requests on Expired contracts. Status transitions work.	15 points - Robust validation logic (Server-side & Client-side). - Edge cases handled gracefully. - Validation logic is separated from Controllers.	7 / 15
Criterion Feedback No logging in to the system					

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4. File Handling Mechanism PDF Uploads.	2 points • Feature missing. • Files not saved.	5 points • File upload works but saves to DB (blob) instead of disk. No extension validation	7 points • Files save to server folder. • Path stored in DB. Basic validation.	10 points • Robust handling (UUID naming to prevent overwrites). • Strict validation (PDF only). Download link works. 	10 / 10
5. External API Integration Currency conversion logic.	4 points • No API integration. • Code blocks thread (.Result).	7 points • Connects to API. Logic is buggy or calculation is incorrect.	11 points • HttpClient used correctly with Async/Await. • Data applied to user input correctly.	15 points • Flawless integration. • Includes error handling (try/catch) if API is down. Math is precise. 	15 / 15
6. Unit Testing: Setup & Coverage Test Project configuration.	4 points • No Test project. Tests fail to build.	7 points • No Test project. Tests fail to build.	11 points • Tests cover Currency Math AND File Validation. Most tests pass.	15 points • Comprehensive test suite. • Mocks used if necessary. • All tests pass (Green). • Automated in GitHub Action 	15 / 15

Criteria	Does not meet the required standard	Meets the required standard	Partially exceeds the required standard	Greatly exceeds the required standard	Criterion Score
7. Unit Testing: Quality Depth of testing.	4 points Tests irrelevant.	7 points Happy path testing only (Expected input).	11 points Tests check for some failures (Invalid input).	15 points - Tests cover edge cases (0 value, .exe file, null inputs). - Demonstrates TDD principles.	15 / 15

Total	87 / 100
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Overall Score

Level 4 ✓

Level 3

Level 2

Level 1